Stat 565, HW 7

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In [29]:
```
import warnings
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import norm, t

from statsmodels.tsa.arima_process import arma_generate_sample
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.stats.diagnostic import acorr_ljungbox

warnings.filterwarnings('ignore')
```

Question 1 set up

In [30]:
```
# Simulate 400 observations from:
# (1 - 0.8B)(Δy_t - 1.6) = (1 + 0.6B)a_t, a_t ~ N(0, 1.5^2)

n = 400
sigma_a = 1.5

ar_poly = np.array([1, -0.8])
ma_poly = np.array([1, 0.6])

w = arma_generate_sample(ar=ar_poly, ma=ma_poly, nsample=n, scale=sigma_a)
x = 1.6 + w
y = np.cumsum(x)

sim_df = pd.DataFrame({'x_diff': x, 'y_level': y})
sim_df.head()
```

Out[30]:

	x_diff	y_level
0	3.641703	3.641703
1	4.532914	8.174617
2	4.395722	12.570340
3	3.491505	16.061845
4	1.003871	17.065716

Question 1 (1)

In [31]:
```
# Method (1): Fit ARIMA(1,1,1) to y_t directly (no exog)
fit_m1 = ARIMA(y, order=(1, 1, 1), trend='n').fit()
fc_m1 = fit_m1.get_forecast(steps=12)

pred_m1 = pd.Series(fc_m1.predicted_mean, index=np.arange(1, 13), name='Meth
pred_m1
```

Out[31]:
```
1     620.271047
2     623.795434
3     626.695530
4     629.081917
5     631.045593
6     632.661432
7     633.991051
8     635.085147
9     635.985441
10    636.726262
11    637.335857
12    637.837472
Name: Method1_yhat, dtype: float64
```

Question 1 (2)

In [32]:
```
# Method (2): Fit ARMA(1,1) to x_t = Δy_t, then map forecasts back to y_t
fit_m2 = ARIMA(x, order=(1, 0, 1), trend='c').fit()
fc_x_m2 = fit_m2.get_forecast(steps=12).predicted_mean

y_last = y[-1]
pred_m2 = y_last + np.cumsum(fc_x_m2)
pred_m2 = pd.Series(pred_m2, index=np.arange(1, 13), name='Method2_yhat')
pred_m2
```

Out[32]:
```
1     620.390946
2     624.211082
3     627.569094
4     630.560692
5     633.261762
6     635.732475
7     638.020540
8     640.163783
9     642.192199
10    644.129569
11    645.994750
12    647.802692
Name: Method2_yhat, dtype: float64
```

Question 1 (3)

In [33]:
```
# Method (3): ARIMA(1,1,1) on y_t with exogenous time trend (xreg analogue)
t = np.arange(1, n + 1)
fit_m3 = ARIMA(y, order=(1, 1, 1), exog=t, trend='n').fit()

future_t = np.arange(n + 1, n + 13)
fc_m3 = fit_m3.get_forecast(steps=12, exog=future_t)
```

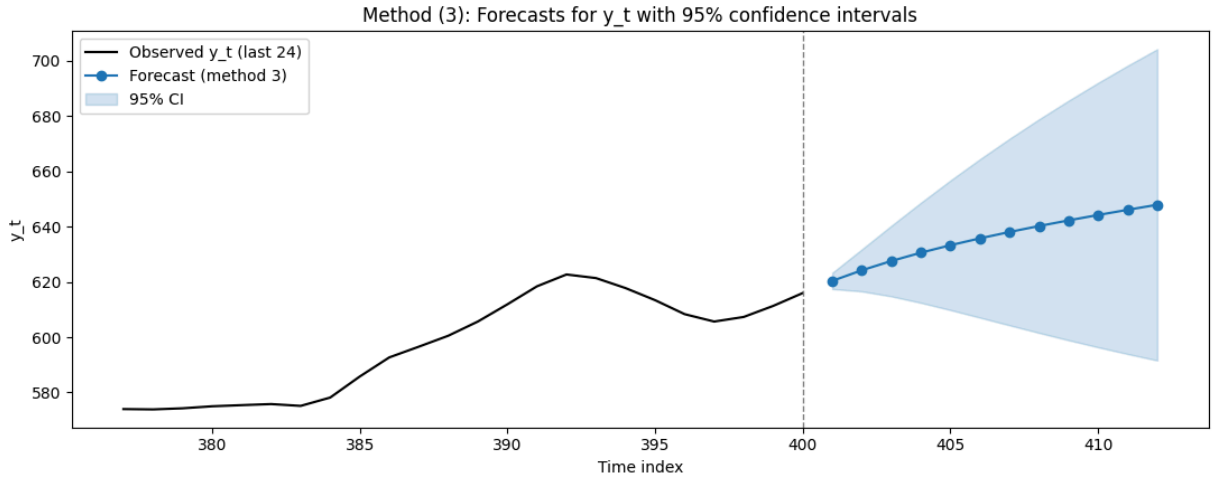
```
pred_m3 = pd.Series(fc_m3.predicted_mean, index=np.arange(1, 13), name='Meth
se_m3 = pd.Series(fc_m3.se_mean, index=np.arange(1, 13), name='Method3_se')

comp_p1 = pd.concat([pred_m1, pred_m2, pred_m3], axis=1)
comp_p1
```

Out[33]:

	Method1_yhat	Method2_yhat	Method3_yhat
1	620.271047	620.390946	620.393742
2	623.795434	624.211082	624.219954
3	626.695530	627.569094	627.586389
4	629.081917	630.560692	630.588069
5	631.045593	633.261762	633.300378
6	632.661432	635.732475	635.783118
7	633.991051	638.020540	638.083735
8	635.085147	640.163783	640.239868
9	635.985441	642.192199	642.281377
10	636.726262	644.129569	644.231951
11	637.335857	645.994750	646.110383
12	637.837472	647.802692	647.931584

In [34]:

```
# Plot last 24 observations, 12-step forecasts, and 95% CI for method (3)
last_24_idx = np.arange(n - 23, n + 1)
future_idx = np.arange(n + 1, n + 13)

ci_m3 = fc_m3.conf_int(alpha=0.05)
if isinstance(ci_m3, np.ndarray):
    ci_low = ci_m3[:, 0]
    ci_high = ci_m3[:, 1]
else:
    ci_low = ci_m3.iloc[:, 0].values
    ci_high = ci_m3.iloc[:, 1].values

plt.figure(figsize=(11, 4.5))
plt.plot(last_24_idx, y[-24:], label='Observed y_t (last 24)', color='black'
plt.plot(future_idx, pred_m3.values, marker='o', label='Forecast (method 3)'
plt.fill_between(future_idx, ci_low, ci_high, color='tab:blue', alpha=0.2, l
plt.axvline(n, color='gray', linestyle='--', linewidth=1)
plt.title('Method (3): Forecasts for y_t with 95% confidence intervals')
plt.xlabel('Time index')
plt.ylabel('y_t')
plt.legend()
plt.tight_layout()
plt.show()

pd.concat([pred_m3, se_m3], axis=1).head()
```



Out[34]:

	Method3_yhat	Method3_se
1	620.393742	1.495473
2	624.219954	3.883968
3	627.586389	6.523950
4	630.588069	9.249386
5	633.300378	11.971062

Question 1 (4)

In [35]:

```
# Check which methods match numerically
check_df = comp_p1.copy()
check_df['abs_diff_2_vs_3'] = np.abs(check_df['Method2_yhat'] - check_df['Me
check_df['abs_diff_1_vs_2'] = np.abs(check_df['Method1_yhat'] - check_df['Me
check_df['abs_diff_1_vs_3'] = np.abs(check_df['Method1_yhat'] - check_df['Me

display(check_df)

print('Max |Method2 - Method3| =', check_df['abs_diff_2_vs_3'].max())
print('Max |Method1 - Method2| =', check_df['abs_diff_1_vs_2'].max())
print('Max |Method1 - Method3| =', check_df['abs_diff_1_vs_3'].max())
```

	Method1_yhat	Method2_yhat	Method3_yhat	abs_diff_2_vs_3	abs_diff_1_vs_2	ab
1	620.271047	620.390946	620.393742	0.002796	0.119899	
2	623.795434	624.211082	624.219954	0.008872	0.415648	
3	626.695530	627.569094	627.586389	0.017295	0.873564	
4	629.081917	630.560692	630.588069	0.027378	1.478774	
5	631.045593	633.261762	633.300378	0.038616	2.216170	
6	632.661432	635.732475	635.783118	0.050643	3.071043	
7	633.991051	638.020540	638.083735	0.063196	4.029489	
8	635.085147	640.163783	640.239868	0.076085	5.078636	
9	635.985441	642.192199	642.281377	0.089178	6.206758	
10	636.726262	644.129569	644.231951	0.102381	7.403308	
11	637.335857	645.994750	646.110383	0.115633	8.658893	
12	637.837472	647.802692	647.931584	0.128892	9.965220	

```
Max |Method2 - Method3| = 0.12889194078650235
Max |Method1 - Method2| = 9.965219735963387
Max |Method1 - Method3| = 10.09411167674989
```

Methods (2) and (3) are essentially the same (with a tiny numerical difference). Method (1) is different because no exogenous drift/trend term is included when fitting y_t directly

Problem 2

In [36]:
```
# Read GE daily OHLC data
ge_path = Path('d-geohlc.txt')
ge = pd.read_csv(ge_path, sep=r'\s+', header=None)
ge.columns = ['year', 'month', 'day', 'open', 'high', 'low', 'close', 'volum
ge['date'] = pd.to_datetime(ge[['year', 'month', 'day']])
ge = ge.sort_values('date').reset_index(drop=True)
ge.head()
```

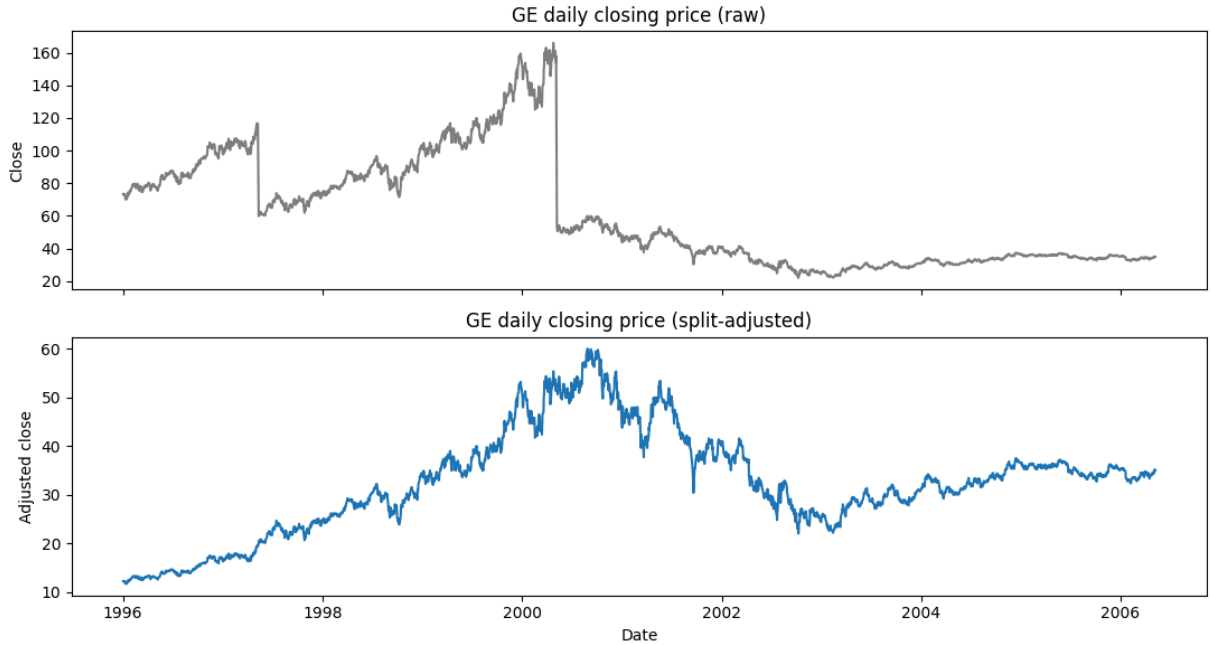
Out[36]:

	year	month	day	open	high	low	close	volume	adj_close	date
0	1996	1	2	71.37	73.62	71.25	73.50	15397800	10.01	1996-01-02
1	1996	1	3	73.75	74.37	73.25	73.62	18469200	10.02	1996-01-03
2	1996	1	4	73.75	74.25	71.25	72.75	18659400	9.91	1996-01-04
3	1996	1	5	72.50	72.87	71.50	72.87	11625000	9.92	1996-01-05
4	1996	1	8	73.25	73.62	72.37	73.37	4343400	9.99	1996-01-08

Question 2 (1)

In [37]:

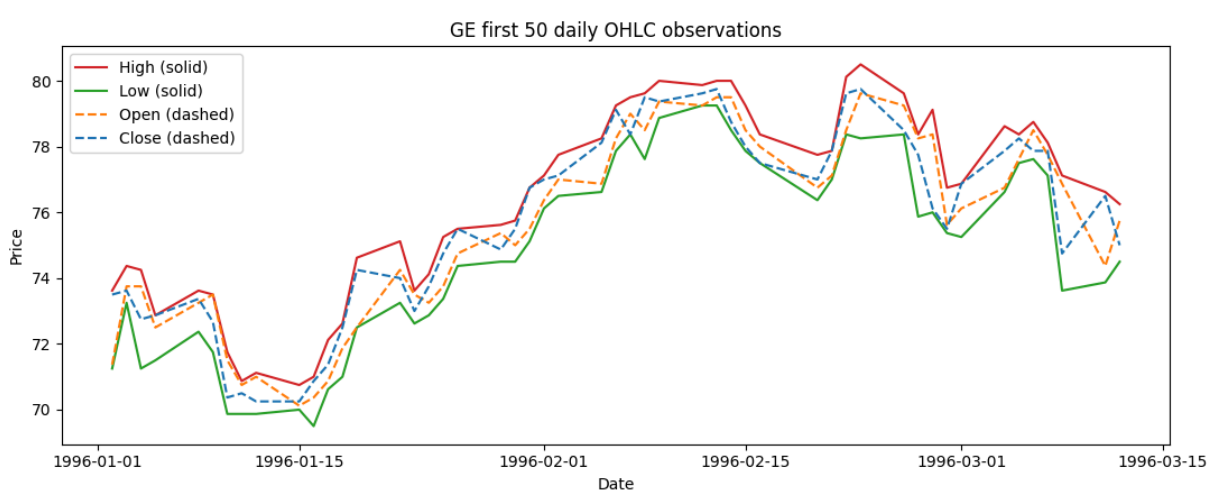
```
# Plot raw close and split-adjusted close
split1 = pd.Timestamp('1997-05-09')
split2 = pd.Timestamp('2000-05-05')

ge['close_adj_split'] = ge['close'].copy()
ge.loc[ge['date'] <= split1, 'close_adj_split'] = ge.loc[ge['date'] <= split
ge.loc[(ge['date'] > split1) & (ge['date'] <= split2), 'close_adj_split'] =
# instructions said to divide before split 1 and not include split2, but it

fig, axes = plt.subplots(2, 1, figsize=(11, 6), sharex=True)
axes[0].plot(ge['date'], ge['close'], color='tab:gray')
axes[0].set_title('GE daily closing price (raw)')
axes[0].set_ylabel('Close')

axes[1].plot(ge['date'], ge['close_adj_split'], color='tab:blue')
axes[1].set_title('GE daily closing price (split-adjusted)')
axes[1].set_ylabel('Adjusted close')
axes[1].set_xlabel('Date')

plt.tight_layout()
plt.show()
```



The raw close series has two visible jumps at split dates. After adjustment, those jumps largely disappear and the long-run movement is much more interpretable

Question 2 (2)

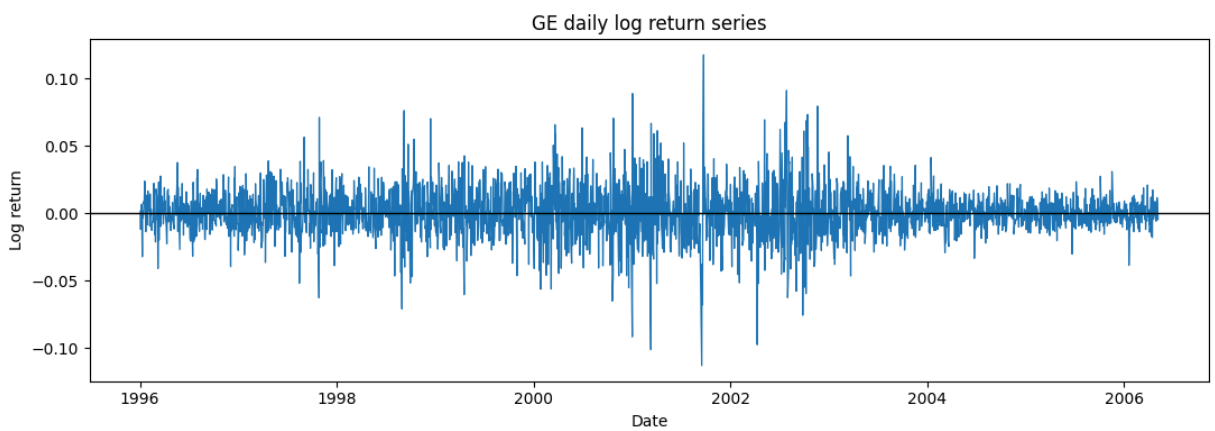
In [38]:
```
# First 50 observations of open/high/low/close on one figure
first50 = ge.iloc[:50]

plt.figure(figsize=(11, 4.5))
plt.plot(first50['date'], first50['high'], color='tab:red', linestyle='-', l
plt.plot(first50['date'], first50['low'], color='tab:green', linestyle='-',
plt.plot(first50['date'], first50['open'], color='tab:orange', linestyle='--
plt.plot(first50['date'], first50['close'], color='tab:blue', linestyle='--'
plt.title('GE first 50 daily OHLC observations')
plt.xlabel('Date')
plt.ylabel('Price')
plt.legend(loc='best')
plt.tight_layout()
plt.show()
```



As expected, daily high and low form an envelope around open and close. Open/close track inside that band with similar short-run dynamics

Question 2 (3)

In [39]:
```
# Daily log return using split-adjusted close
ge['log_close_adj'] = np.log(ge['close_adj_split'])
ge['ret'] = ge['log_close_adj'].diff()
ret = ge['ret'].dropna()

plt.figure(figsize=(11, 4))
plt.plot(ge.loc[ret.index, 'date'], ret, color='tab:blue', linewidth=0.9)
plt.axhline(0, color='black', linewidth=1)
plt.title('GE daily log return series')
plt.xlabel('Date')
plt.ylabel('Log return')
plt.tight_layout()
plt.show()

ret.describe()
```



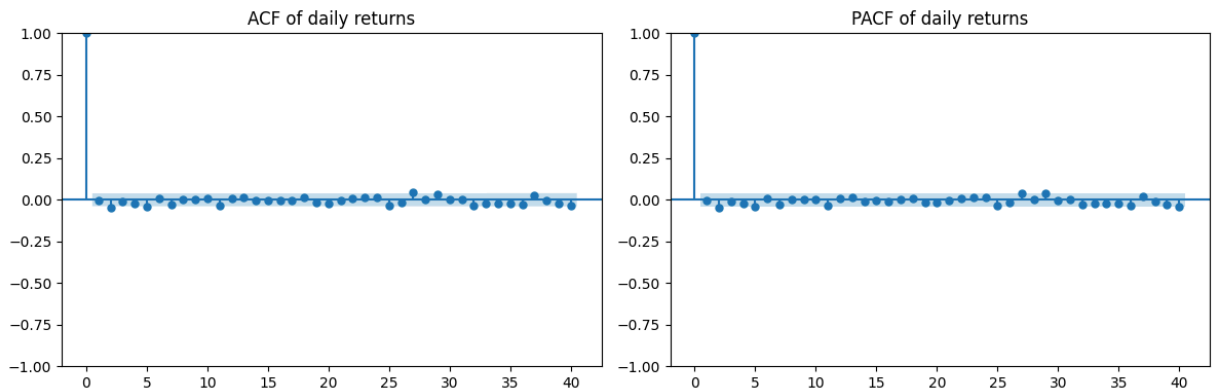
Out[39]:
```
count    2604.000000
mean        0.000403
std         0.018450
min        -0.112871
25%        -0.009468
50%         0.000000
75%         0.010488
max         0.117428
Name: ret, dtype: float64
```

The return series fluctuates around zero with periods of high and low volatility, suggesting volatility clustering

Question 2 (4)

In [40]:
```
# ACF and PACF of returns
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
plot_acf(ret, lags=40, ax=axes[0])
plot_pacf(ret, lags=40, ax=axes[1], method='ywm')
```

```
axes[0].set_title('ACF of daily returns')
axes[1].set_title('PACF of daily returns')
plt.tight_layout()
plt.show()
```



The return autocorrelations are small overall, with only mild short-lag dependence. This suggests a low-order ARMA model for the mean is reasonable -- we find this to be true in the next step with ARMA(1,2) being the best fit

Question 2 (5)

In [41]:
```
# Fit candidate ARMA(p,q) models and choose by AIC + Ljung-Box
candidates = []
arma_fits = {}

for p in range(0, 4):
    for q in range(0, 4):
        if p == 0 and q == 0:
            continue
        try:
            fit = ARIMA(ret, order=(p, 0, q), trend='c').fit()
            resid = fit.resid
            lb_p = float(acorr_ljungbox(resid, lags=[20], model_df=(p+q), re
            candidates.append({'model': f'ARMA({p},{q})', 'p': p, 'q': q, 'A
            arma_fits[(p, q)] = fit
        except Exception:
            pass

arma_table = pd.DataFrame(candidates).sort_values('AIC').reset_index(drop=Tr
display(arma_table.head(10))

best = arma_table.iloc[0]
best_p, best_q = int(best['p']), int(best['q'])
best_name = best['model']
best_fit = arma_fits[(best_p, best_q)]
mean_resid = best_fit.resid

print('Selected mean model:', best_name)
print(best_fit.summary())
```

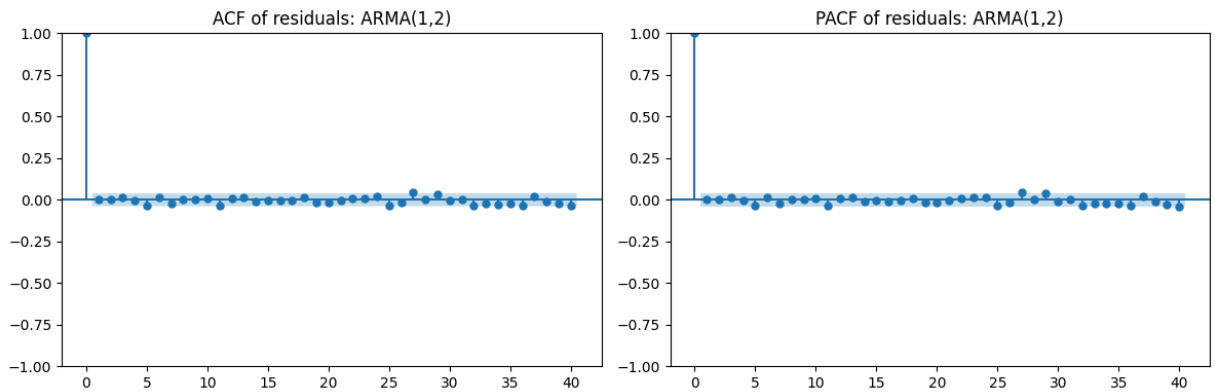
	model	p	q	AIC	LjungBox_p_lag20
0	ARMA(1,2)	1	2	-13404.686981	0.742576
1	ARMA(2,2)	2	2	-13403.809155	0.750677
2	ARMA(0,2)	0	2	-13403.433165	0.607820
3	ARMA(2,0)	2	0	-13403.118965	0.589458
4	ARMA(0,3)	0	3	-13401.944315	0.569832
5	ARMA(3,3)	3	3	-13401.924289	0.785452
6	ARMA(2,1)	2	1	-13401.726427	0.556740
7	ARMA(2,3)	2	3	-13401.601090	0.668680
8	ARMA(3,0)	3	0	-13401.377000	0.533584
9	ARMA(1,3)	1	3	-13400.317449	0.523577

```
Selected mean model: ARMA(1,2)
                               SARIMAX Results
==============================================================================
==
Dep. Variable:                    ret   No. Observations:                 26
04
Model:                 ARIMA(1, 0, 2)   Log Likelihood                6707.3
43
Date:                Tue, 21 Apr 2026   AIC                         -13404.6
87
Time:                        15:48:54   BIC                         -13375.3
63
Sample:                             0   HQIC                        -13394.0
63
                               - 2604
Covariance Type:                  opg
==============================================================================
==
                 coef    std err          z      P>|z|      [0.025      0.97
5]
------------------------------------------------------------------------------
--
const          0.0004      0.000      1.214      0.225      -0.000       0.0
01
ar.L1          0.5769      0.163      3.549      0.000       0.258       0.8
96
ma.L1         -0.5840      0.164     -3.568      0.000      -0.905      -0.2
63
ma.L2         -0.0454      0.017     -2.752      0.006      -0.078      -0.0
13
sigma2         0.0003   5.78e-06     58.656      0.000       0.000       0.0
00
===============================================================================
=======
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):
1281.29
Prob(Q):                              1.00   Prob(JB):
0.00
Heteroskedasticity (H):               0.50   Skew:
-0.02
Prob(H) (two-sided):                  0.00   Kurtosis:
6.44
===============================================================================
=======

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (compl
ex-step).
```

I pick the ARMA model with the smallest AIC among stable fitted candidates -- ARMA(1, 2) -- and check residual whiteness with Ljung-Box

Question 2 (6)

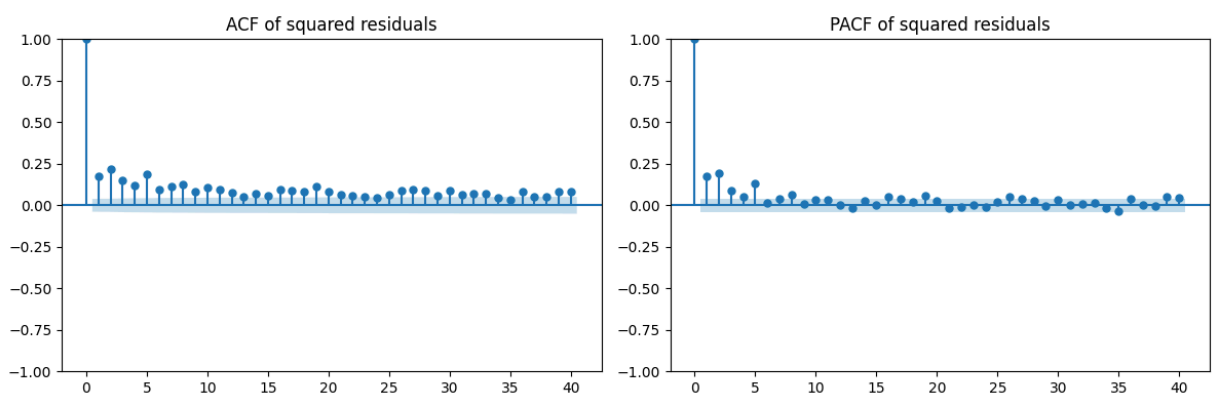
In [42]:
```
# ACF/PACF of ARMA residuals
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
plot_acf(mean_resid, lags=40, ax=axes[0])
plot_pacf(mean_resid, lags=40, ax=axes[1], method='ywm')
axes[0].set_title(f'ACF of residuals: {best_name}')
axes[1].set_title(f'PACF of residuals: {best_name}')
plt.tight_layout()
plt.show()
```



Residual autocorrelation is weak after fitting the ARMA mean model, indicating the mean dynamics are mostly absorbed

Question 2 (7)

In [43]:
```
# ACF/PACF of squared ARMA residuals
sq_resid = mean_resid**2

fig, axes = plt.subplots(1, 2, figsize=(12, 4))
plot_acf(sq_resid, lags=40, ax=axes[0])
plot_pacf(sq_resid, lags=40, ax=axes[1], method='ywm')
axes[0].set_title('ACF of squared residuals')
axes[1].set_title('PACF of squared residuals')
plt.tight_layout()
plt.show()
```



The squared residual autocorrelations are clearly more pronounced than in raw residuals, which is evidence of conditional heteroskedasticity and suggests that we should do GARCH-type variance modeling

Question 2 (8)

In [44]:
```
# Fit GARCH models on ARMA residuals (normal first), then compare with t err
try:
    from arch import arch_model
except ImportError as e:
    raise ImportError("Please install the 'arch' package before running Ques

# Scale residuals for numerical stability in arch fitting
resid_scale = mean_resid * 100

garch_rows = []
garch_fits = {}
for p in [1, 2]:
    for q in [1, 2]:
        try:
            gfit = arch_model(resid_scale, mean='Zero', vol='GARCH', p=p, q=
            garch_rows.append({'model': f'GARCH({p},{q})-Normal', 'p': p, 'c
            garch_fits[(p, q, 'normal')] = gfit
        except Exception:
            pass

garch_table = pd.DataFrame(garch_rows).sort_values('AIC').reset_index(drop=T
display(garch_table)

best_garch = garch_table.iloc[0]
bp, bq = int(best_garch['p']), int(best_garch['q'])
fit_garch_norm = garch_fits[(bp, bq, 'normal')]
fit_garch_t = arch_model(resid_scale, mean='Zero', vol='GARCH', p=bp, q=bq,

print('Selected normal model by AIC:', f'GARCH({bp},{bq})-Normal')
print('\nNormal errors fit summary:')
print(fit_garch_norm.summary())
print('\nStudent-t errors fit summary:')
print(fit_garch_t.summary())

if fit_garch_t.aic < fit_garch_norm.aic:
    final_garch = fit_garch_t
    final_dist = 't'
else:
    final_garch = fit_garch_norm
    final_dist = 'normal'

print('\nFinal variance model:', f'GARCH({bp},{bq}) with {final_dist} errors
```

	model	p	q	AIC	BIC
0	GARCH(1,1)-Normal	1	1	9936.776443	9954.370855
1	GARCH(1,2)-Normal	1	2	9937.782933	9961.242149
2	GARCH(2,1)-Normal	2	1	9938.776444	9962.235660
3	GARCH(2,2)-Normal	2	2	9940.127027	9969.451047

```
Selected normal model by AIC: GARCH(1,1)-Normal

Normal errors fit summary:
                       Zero Mean - GARCH Model Results
==============================================================================
==
Dep. Variable:                   None   R-squared:                       0.0
00
Mean Model:                 Zero Mean   Adj. R-squared:                  0.0
00
Vol Model:                      GARCH   Log-Likelihood:               -4965.
39
Distribution:                  Normal   AIC:                           9936.
78
Method:            Maximum Likelihood   BIC:                           9954.
37
                                        No. Observations:                 26
04
Date:                Tue, Apr 21 2026   Df Residuals:                     26
04
Time:                        15:48:54   Df Model:
0
                              Volatility Model
==============================================================================
=
                 coef    std err          t      P>|t|       95.0% Conf. In
t.
------------------------------------------------------------------------------
-
omega      5.0060e-03  5.979e-03      0.837      0.402 [-6.713e-03,1.673e-0
2]
alpha[1]       0.0339  1.267e-02      2.676  7.446e-03  [9.078e-03,5.876e-0
2]
beta[1]        0.9651  1.379e-02     69.991      0.000      [  0.938,  0.99
2]
==============================================================================
=

Covariance estimator: robust

Student-t errors fit summary:
                         Zero Mean - GARCH Model Results
==============================================================================
========
Dep. Variable:                         None   R-squared:
0.000
Mean Model:                       Zero Mean   Adj. R-squared:
0.000
Vol Model:                           GARCH   Log-Likelihood:
-4924.36
Distribution:      Standardized Student's t   AIC:
9856.72
Method:                  Maximum Likelihood   BIC:
9880.18
                                              No. Observations:
2604
```

```
Date:                      Tue, Apr 21 2026   Df Residuals:
2604
Time:                              15:48:54   Df Model:
0
                              Volatility Model
==============================================================================
=
                 coef    std err          t      P>|t|       95.0% Conf. In
t.
------------------------------------------------------------------------------
-
omega      5.7681e-03  4.623e-03      1.248      0.212 [-3.293e-03,1.483e-0
2]
alpha[1]       0.0370  1.030e-02      3.594  3.255e-04  [1.683e-02,5.719e-0
2]
beta[1]        0.9616  1.076e-02     89.369      0.000      [  0.941,  0.98
3]
                              Distribution
========================================================================
                 coef    std err          t      P>|t|  95.0% Conf. Int.
------------------------------------------------------------------------
nu             9.0936      1.499      6.067  1.302e-09 [  6.156, 12.031]
========================================================================

Covariance estimator: robust

Final variance model: GARCH(1,1) with t errors
```

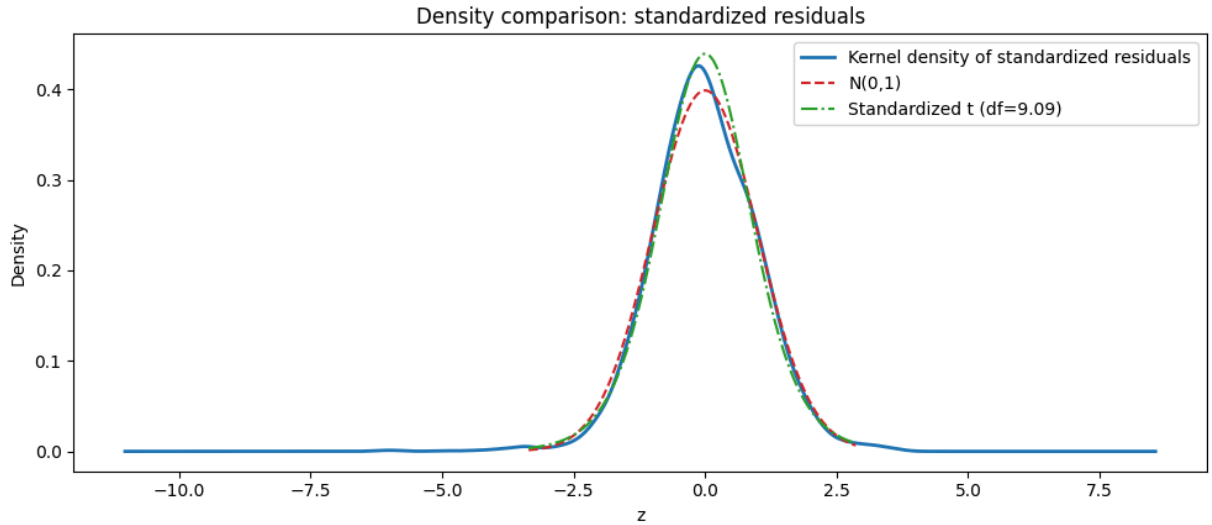
I start from normal-error GARCH(p,q) candidates and select the best by information criteria, then compare against the same order with t errors. The final model Garch(1, 1) has the better fit and sensible parameter estimates

Question 2 (9)

In [45]:

```
# Standardized residual density vs Normal (and standardized t if selected)
from scipy.stats import t as student_t

std_resid = pd.Series(final_garch.std_resid).dropna()

xgrid = np.linspace(std_resid.quantile(0.005), std_resid.quantile(0.995), 40

plt.figure(figsize=(10, 4.5))
std_resid.plot(kind='kde', linewidth=2, label='Kernel density of standardize
plt.plot(xgrid, norm.pdf(xgrid, loc=0, scale=1), '--', label='N(0,1)', color

if final_dist == 't':
    nu = float(final_garch.params.get('nu', np.nan))
    if np.isfinite(nu) and nu > 2:
        scale = np.sqrt((nu - 2) / nu)
        plt.plot(xgrid, student_t.pdf(xgrid / scale, df=nu) / scale, '-.', c
                 label=f'Standardized t (df={nu:.2f})')

plt.title('Density comparison: standardized residuals')
plt.xlabel('z')
plt.ylabel('Density')
```

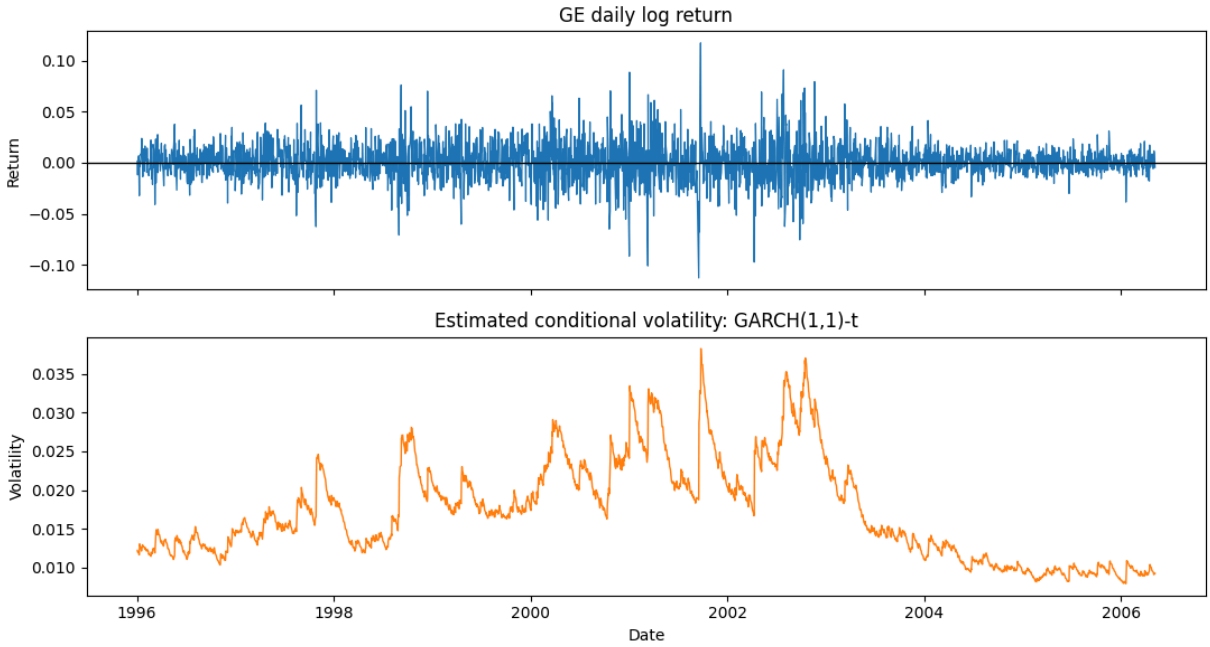
```
plt.legend()
plt.tight_layout()
plt.show()
```



The residual density is compared to N(0,1), and to standardized t if the t-GARCH model is selected. Any heavier tails in the empirical density relative to normal indicate non-Gaussian shock behavior

Question 2 (10)

In [46]:
```
# Plot return series and estimated volatility on same page
cond_vol = pd.Series(final_garch.conditional_volatility, index=mean_resid.ir

fig, axes = plt.subplots(2, 1, figsize=(11, 6), sharex=True)
axes[0].plot(ge.loc[mean_resid.index, 'date'], ret.loc[mean_resid.index], cc
axes[0].axhline(0, color='black', linewidth=1)
axes[0].set_title('GE daily log return')
axes[0].set_ylabel('Return')

axes[1].plot(ge.loc[mean_resid.index, 'date'], cond_vol, color='tab:orange',
axes[1].set_title(f'Estimated conditional volatility: GARCH({bp},{bq})-{fina
axes[1].set_ylabel('Volatility')
axes[1].set_xlabel('Date')

plt.tight_layout()
plt.show()
```



Estimated volatility rises during turbulent periods and falls during calm periods, consistent with volatility clustering visible in the return plot